

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

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1.Preprocessing and Feature Detection

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent. Analyze the issue and explain how preprocessing improves feature detection performance.

PREPROCESSING AND FEATURE DETECTION

A vision system may capture images under different lighting conditions such as bright sunlight, low light, shadows, or uneven illumination. These changes affect the intensity, contrast, and visibility of image features. As a result, feature detection may produce different or unreliable key points from similar images.

Preprocessing is therefore performed before feature detection to improve image quality and make important features more consistent.

First, grayscale conversion can be used to simplify the image by removing unnecessary color information. Next, noise reduction using filters such as Gaussian or median filtering removes unwanted noise that may cause false feature points.

For example, if an object is captured once in bright light and once in low light, some corners and edges may be clearly visible in the first image but weak in the second. After preprocessing, the contrast of weak regions is improved, allowing feature detectors such as SIFT, ORB, or Harris to identify more stable key points.

Without Preprocessing	With Preprocessing
Uneven brightness	More uniform intensity
Low-contrast features	Enhanced features
More noise and false points	Reduced noise
Inconsistent key points	Stable and repeatable key points
Poor matching accuracy	Improved matching accuracy

Thus, preprocessing acts as an important step before feature detection. It reduces the effects of illumination, noise, and contrast variations, resulting in more reliable and repeatable features. Therefore, techniques such as grayscale conversion, noise filtering, CLAHE, and normalization can significantly improve feature detection and subsequent image matching performance.

2. Edge vs Corner Detection

In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching.

Analyze why edge detection is insufficient and justify the use of corner detection in this case.

EDGE VS CORNER DETECTION

In the given scenario, the image contains straight edges and textured regions. Although edge detection can identify object boundaries, it may fail to provide reliable key points for image matching. This is because an edge represents an intensity change mainly in one direction and may extend over a long continuous region. Many points along the same edge can look similar, making it difficult to identify a unique matching point.

For example, a straight line may contain hundreds of edge pixels, but these pixels have almost the same local structure. Therefore, edge detection alone cannot determine which specific point corresponds to the same point in another image.

Corner detection is more suitable because corners are points where intensity changes significantly in two or more directions. This makes them more distinctive and easier to locate accurately. Common methods include Harris Corner Detection, Shi-Tomasi, and FAST.

Edge Detection	Corner Detection
Detects lines and boundaries	Detects distinctive key points
Changes mainly in one direction	Changes in multiple directions
Points may not be unique	Points are more distinctive
Less suitable for matching	Better suited for matching
Example: Canny	Example: Harris

Thus, edge detection is useful for identifying object boundaries, but it is insufficient when unique points are required for feature matching. Corner detection provides stable, localized, and distinctive key points, improving the accuracy of image representation and comparison. Therefore, Harris or Shi-Tomasi corner detection should be used along with or instead of edge detection for reliable feature matching.

6. Preprocessing Impact on Edge Detection

A system performs edge detection on raw images containing noise and uneven illumination, resulting in fragmented edges.

Analyze the problem and explain how preprocessing improves edge detection accuracy and continuity.

PREPROCESSING IMPACT ON EDGE DETECTION

In the given scenario, edge detection is performed directly on raw images containing noise and uneven illumination. This causes the detected edges to become fragmented, weak, or discontinuous. Noise creates unwanted intensity changes, which may be incorrectly detected as edges, while uneven lighting can make some actual edges too weak to detect.

Preprocessing helps improve the quality and continuity of detected edges. Grayscale conversion simplifies the image by using intensity information. Gaussian or median filtering reduces noise and prevents false edges. Contrast enhancement improves the visibility of weak edges, especially in poorly illuminated regions. Histogram Equalization or CLAHE can reduce the effect of uneven illumination and produce more uniform contrast.

After preprocessing, an edge detector such as Canny can identify actual boundaries more accurately. The Canny method uses smoothing, gradient calculation, non-maximum suppression, and thresholding to produce thin and connected edges.

Without Preprocessing	With Preprocessing
High noise	Reduced noise
False edges	Fewer false edges
Fragmented edges	More continuous edges
Weak edges	Stronger visible edges
Lower detection accuracy	Higher detection accuracy

Conclusion: Preprocessing improves edge detection by reducing noise, correcting illumination variations, and enhancing contrast. Therefore, a pipeline such as Grayscale → Gaussian Filtering → CLAHE/Contrast Enhancement → Canny Edge Detection provides more accurate, continuous, and reliable edges.

8. Hough Transform for Shape Detection

A system uses Hough Transform to detect circular shapes but fails in complex backgrounds. Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

HOUGH TRANSFORM FOR SHAPE DETECTION

The Hough Transform is a popular computer vision technique used to detect geometric shapes such as circles, lines, and ellipses. For circle detection, it identifies possible circles based on their center coordinates and radius. However, in complex backgrounds containing noise, multiple edges, shadows, and unwanted objects, the system may detect false circles or miss the actual circular shape.

The main limitation is that Hough Transform depends heavily on the quality of the detected edges. Noise and unnecessary edges increase the number of possible candidates, making detection less accurate. It can also be computationally expensive when a large range of circle radii is considered.

Improving Detection

Preprocessing can significantly improve the results. Grayscale conversion simplifies the image, while Gaussian or median filtering reduces noise. Thresholding or edge detection using Canny can remove irrelevant background information and provide clearer boundaries.

Parameter tuning is also important. In circle detection, parameters such as minimum and maximum radius, accumulator threshold, minimum distance between detected centers, and edge-detection thresholds should be selected according to the expected object size and image quality.

Problem	Improvement
Image noise	Gaussian/Median filtering
Complex background	Thresholding or ROI selection
Weak edges	Contrast enhancement
False circles	Increase accumulator threshold
Wrong circle size	Set suitable radius range
Multiple nearby circles	Adjust minimum center distance

Conclusion: Hough Transform is effective for detecting circular shapes, but its performance decreases in noisy and complex scenes. Combining preprocessing, ROI selection, Canny edge detection, and proper parameter tuning improves detection accuracy, reduces false detections, and makes circle detection more reliable.

11. Edge Detection in Low Contrast Images

An image has low contrast, causing edge detection algorithms to fail.

Analyze the problem and explain how preprocessing enhances edge detection results.

EDGE DETECTION IN LOW CONTRAST IMAGES

In a low-contrast image, the difference in intensity between an object and its background is very small. As a result, edge detection algorithms such as Canny, Sobel, or Prewitt may fail to identify important boundaries. The detected edges may be weak, incomplete, or missing completely.

The main problem is that edge detectors depend on intensity gradients. When the contrast is low, these gradients are small and may not exceed the detection threshold. Noise can also become significant compared with the actual edge information, resulting in inaccurate detection.

Role of Preprocessing

Preprocessing improves edge detection by increasing the difference between important regions and reducing unwanted variations.

- **Grayscale conversion** simplifies the image and focuses on intensity information.
- **Histogram Equalization** improves the overall contrast and makes weak edges more visible.
- **CLAHE** enhances local contrast, making it particularly useful for images with uneven illumination.
- **Gaussian filtering** reduces noise before edge detection and prevents false edges.
- **Image normalization** adjusts intensity values to a suitable range.

Before Preprocessing	After Preprocessing
Low contrast	Improved contrast
Weak edges	Stronger edges
Missing boundaries	Clearer boundaries
More detection errors	Better accuracy

Conclusion: Preprocessing enhances low-contrast images by improving contrast, reducing noise, and strengthening intensity differences. A suitable pipeline such as Grayscale → Noise Reduction → CLAHE/Histogram Equalization → Canny Edge Detection produces clearer and more continuous edges, improving overall edge detection performance.